



Scoreboard		Low-end PCI Express Graphics Cards						
SEPTEMBER 2005		SEPTEMBER 2005						
BRAND	Gainward	Gigabyte	Gigabyte	MaxForce	MaxForce	MSI	PowerColor	XFx
Model	GeForce 6200 256 MB	GeForce 6200 128 MB	X300 SE 128 MB	GeForce 6200 128 MB	X300 SE 128 MB	GeForce 6200 128 MB	X300 256 MB	GeForce 6200 256 MB
Chipset	NVIDIA GeForce 6200TC	NVIDIA GeForce 6200TC	ATI X300 SE	GeForce 6200TC	ATI X300 SE	NVIDIA GeForce 6200TC	ATI X300	NVIDIA GeForce 6200TC
Features	28.70	20.65	21.35	25.55	21.35	21.35	25.90	27.30
Dual Monitor Support	4	6	4	4	4	4	4	4
Software Bundle	DVI-to-VGA adapter, S-Video cable, Driver CD	DVI-to-VGA adapter, Driver CD	S-Video-to-Composite adapter, Driver CD	DVI-to-VGA adapter, S-Video cable, Driver CD	S-Video-to-Composite adapter, Driver CD	Driver CD	DVI-to-VGA adapter, S-Video cable, Driver CD	S-Video Cable, Driver CD
Specifications								
Memory (MB)	256	128	128	128	128	128	256	256
Manufacturing (nm)	110	110	110	110	110	110	110	110
Core Clock Speed (MHz)	350	350	325	350	325	350	400	350
Effective Speed (MHz)	550	700	400	700	400	400	400	400
Memory Bus Width (bits)	64	32	64	32	64	64	128	64
Max Resolution	2048x1536	2048x1536	2048x1536	2048x1536	2048x1536	2048x1536	2048x1536	2048x1536
Compatibility (DX/OpenGL)	DX 9.0c/ OpenGL 2.0	DX 9.0c/ OpenGL 2.0	DX 9.0b/ OpenGL 2.0	DX 9.0c/ OpenGL 2.0	DX 9.0b/ OpenGL 2.0	DX 9.0c/ OpenGL 2.0	DX 9.0b/ OpenGL 2.0	DX 9.0c/ OpenGL 2.0
Pixel Pipelines	4	4	4	4	4	4	4	4
Pixel Shader/Vertex Shader	3.0/3.0	3.0/3.0	2.0/2.0	3.0/3.0	2.0/2.0	3.0/3.0	2.0/2.0	3.0/3.0
Performance	31.98	25.96	28.10	25.56	26.96	30.59	38.25	27.45
3D - OpenGL								
Doom 3								
Normal (640x480x32) (fps)	41.30	32.00	32.70	32.10	31.10	40.10	48.40	33.40
High (800x600x32) (fps)	29.90	22.30	22.10	22.20	21.20	28.80	32.10	24.00
Max (1024x768x32) (fps)	20.30	15.40	15.00	14.80	14.40	19.70	22.20	16.20
3D - Direct3D								
Half-Life 2								
Normal (640x480x32) (fps)	85.49	77.15	93.89	76.43	91.44	85.30	105.25	82.61
High (800x600x32) (fps)	68.56	60.48	72.25	60.20	71.22	37.93	91.71	63.81
Max (1024x768x32) (fps)	48.13	42.14	48.19	41.65	47.44	47.22	65.07	43.93
FarCry								
Normal (640x480x32) (fps)	96.85	92.91	104.89	90.76	103.45	96.01	113.66	93.53
High (800x600x32) (fps)	83.34	69.93	77.88	68.35	75.78	84.92	106.03	72.10
Max (1024x768x32) (fps)	58.80	47.47	49.42	46.53	47.72	57.99	72.19	49.50
3DMark 03 score								
Low End (800x600x32)	3602	2721	1988	2679	2711	3595	4211	2968
High End (1024x768x32) (fps)	2659	1997	2851	1963	1890	2641	3098	2166
Price Index	14.63	16.51	16.51	17.00	14.82	9.17	9.71	16.51
Price (Rs)	3950	3500	3500	3400	3900	6300	5950	3500
Overall Score (Out of 100)	75.31	63.12	65.96	68.11	63.13	61.12	73.86	71.26

On the other hand, the X600 runs a clear 100 MHz less, and has four pixel pipelines less. The X700 is more in the class of the 6600 GT, but its performance is much lower, as you will see in the performance analysis.

Performance

The 6600 series dominated here. In most game tests, the ATi cards couldn't match the nVidias. None of the ATis posted scores that could match up to the 6600 cards. At higher resolutions, the 6600 non-GT cards and the ATi cards could only manage a maximum of 40 fps. The GTs comfortably cruised above 50 fps in most DirectX games. This translates into a good gameplay experience.

Users who purchase a mid-range video card look for value of money as well as a good, if not excellent, gaming experience. This is what we based our tests on.

The price these cards retail at is generally in the range of Rs 5,000 to Rs 15,000. For a user who wants decent gameplay at resolutions of up to

1024 x 768, a regular non-GT 6600 is more than enough. In most cases, you can turn on a little of the eye-candy in games, but not all of it. The only ATi card we can recommend here is the X700. The other cards were barely able to provide playable frame rates—even at 1024 x 768.



XFx GeForce 6600GT
256 MB

